

# Dialogues from Home to improve the vocabulary in English

1

00:01:28,416 --> 00:01:31,041

Listen to me, please.

2

00:01:31,208 --> 00:01:35,000

You're like me, a homo sapiens,

3

00:01:35,166 --> 00:01:37,500

a wise human.

4

00:01:37,666 --> 00:01:38,875

Life,

5

00:01:39,041 --> 00:01:43,791

a miracle in the universe,  
appeared around 4 billion years ago.

6

00:01:43,958 --> 00:01:48,125

And we humans

only 200,000 years ago.

7

00:01:48,291 --> 00:01:53,791

Yet we have succeeded in disrupting  
the balance so essential to life.

8

00:01:53,958 --> 00:01:57,833

Listen carefully to this  
extraordinary story, which is yours,

9

00:01:58,000 --> 00:02:01,333

and decide  
what you want to do with it.

10

00:02:19,125 --> 00:02:22,000

These are traces of our origins.

11

00:02:22,166 --> 00:02:26,750

At the beginning, our planet  
was no more than a chaos of fire,

12

00:02:26,916 --> 00:02:29,541

a cloud

of agglutinated dust particles,

13

00:02:29,708 --> 00:02:33,458

like so many similar clusters  
in the universe.

14

00:02:33,958 --> 00:02:37,416

Yet this is where  
the miracle of life occurred.

15

00:03:25,541 --> 00:03:28,000

Today, life, our life,

16

00:03:28,166 --> 00:03:31,791

is just a link in a chain  
of innumerable living beings

17

00:03:31,958 --> 00:03:36,500

that have succeeded one another  
on Earth over nearly 4 billion years.

18

00:03:40,708 --> 00:03:42,041

And even today,

19

00:03:42,208 --> 00:03:45,958

new volcanoes continue  
to sculpt our landscapes.

20

00:03:46,125 --> 00:03:49,791

They offer a glimpse of what  
our Earth was like at its birth,

21

00:03:49,958 --> 00:03:52,541

molten rock surging from the depths,

22

00:03:52,708 --> 00:03:58,000

solidifying, cracking, blistering  
or spreading in a thin crust,

23

00:03:58,166 --> 00:04:00,958

before falling dormant for a time.

24

00:04:13,666 --> 00:04:17,541

These wreathes of smoke  
curling from the bowels of the Earth

25

00:04:17,708 --> 00:04:21,041

bear witness

to the Earth's original atmosphere.

26

00:04:21,208 --> 00:04:24,500

An atmosphere devoid of oxygen.

27

00:04:24,666 --> 00:04:27,750

A dense atmosphere,

thick with water vapor,

28

00:04:27,916 --> 00:04:30,208

full of carbon dioxide.

29

00:04:31,083 --> 00:04:32,750

A furnace.

30

00:04:42,875 --> 00:04:44,666

The Earth cooled.

31

00:04:44,916 --> 00:04:49,208

The water vapor condensed

and fell in torrential downpours.

32

00:04:49,375 --> 00:04:53,958

At the right distance from the sun,  
not too far, not too near,

33

00:04:54,125 --> 00:04:57,875

the Earth's perfect balance  
enabled it to conserve water

34

00:04:58,041 --> 00:05:00,125

in liquid form.

35

00:05:00,291 --> 00:05:02,625

The water cut channels.

36

00:05:02,791 --> 00:05:06,625

They are like the veins of a body,  
the branches of a tree,

37

00:05:06,791 --> 00:05:10,833

the vessels of the sap  
that the water gave to the Earth.

38

00:05:22,083 --> 00:05:28,250

The rivers tore minerals from rocks,  
adding them to the oceans' freshwater.

39

00:05:28,791 --> 00:05:32,833

And the oceans became heavy with salt.

40

00:05:50,333 --> 00:05:52,375

Where do we come from?

41

00:05:52,541 --> 00:05:55,666

Where did life  
first spark into being?

42

00:05:57,208 --> 00:05:58,875

A miracle of time,

43

00:05:59,041 --> 00:06:03,041

primitive life forms still exist  
in the globe's hot springs.

44

00:06:03,208 --> 00:06:07,625

They give them their colors.

They're called archeobacteria.

45

00:06:16,458 --> 00:06:18,958

They all feed off the Earth's heat.

46

00:06:19,125 --> 00:06:21,333

All except the cyanobacteria,

47

00:06:21,875 --> 00:06:24,416

or blue-green algae.

48

00:06:24,583 --> 00:06:27,083

They alone have the capacity

to turn to the sun

49

00:06:27,250 --> 00:06:29,791

to capture its energy.

50

00:06:30,833 --> 00:06:36,750

They are a vital ancestor of all

yesterday's and today's plant species.



51

00:06:37,458 --> 00:06:41,791

These tiny bacteria  
and their billions of descendants

52

00:06:41,958 --> 00:06:45,291

changed the destiny of our planet.

53

00:06:45,458 --> 00:06:49,041

They transformed its atmosphere.

54

00:06:53,791 --> 00:06:57,833

What happened to the carbon  
that poisoned the atmosphere?

55

00:06:58,833 --> 00:07:02,958

It's still here,  
imprisoned in the Earth's crust.

56

00:07:05,166 --> 00:07:09,375

Here, there once was a sea,  
inhabited by micro-organisms.

57

00:07:09,708 --> 00:07:13,416

They grew shells by tapping into  
the atmosphere's carbon

58

00:07:13,583 --> 00:07:16,083

now dissolved in the ocean.

59

00:07:16,666 --> 00:07:19,291

These strata  
are the accumulated shells

60

00:07:19,458 --> 00:07:23,750

of those billions and billions  
of micro-organisms.

61

00:07:29,166 --> 00:07:33,041

Thanks to them, the carbon drained  
from the atmosphere

62

00:07:33,208 --> 00:07:35,708

and other life forms could develop.

63

00:07:39,458 --> 00:07:42,916

It is life

that altered the atmosphere.

64

00:07:44,500 --> 00:07:47,375

Plant life fed off the sun's energy,

65

00:07:47,541 --> 00:07:52,166

which enabled it to break apart

the water molecule and take the oxygen.

66

00:07:52,583 --> 00:07:56,000

And oxygen filled the air.

67

00:07:56,583 --> 00:08:00,833

The Earth's water cycle

is a process of constant renewal.

68

00:08:01,541 --> 00:08:04,166

Waterfalls, water vapor,

69

00:08:04,333 --> 00:08:06,250

clouds, rain,

70

00:08:06,708 --> 00:08:09,291

springs, rivers,

71

00:08:09,458 --> 00:08:13,166

seas, oceans, glaciers...

72

00:08:13,333 --> 00:08:15,375

The cycle is never broken.

73

00:08:15,541 --> 00:08:18,958

There's always the same quantity  
of water on Earth.

74

00:08:19,125 --> 00:08:24,208

All the successive species on Earth  
have drunk the same water.

75

00:08:25,125 --> 00:08:28,208

The astonishing matter that is water.

76

00:08:28,375 --> 00:08:30,500

One of the most unstable of all.

77

00:08:30,666 --> 00:08:33,250

It takes a liquid form  
as running water,

78

00:08:33,416 --> 00:08:37,833

gaseous as vapor,  
or solid as ice.

79

00:08:40,666 --> 00:08:44,833

In Siberia, the frozen surfaces  
of the lakes in winter

80

00:08:45,000 --> 00:08:49,250

contain the trace of the forces  
that water deploys when it freezes.

81

00:08:49,416 --> 00:08:51,791

Lighter than water, the ice floats.

82

00:08:51,958 --> 00:08:54,958

It forms a protective mantle

against the cold,

83

00:08:55,125 --> 00:08:57,791

under which life can go on.

84

00:09:33,458 --> 00:09:36,708

The engine of life is linkage.

85

00:09:36,875 --> 00:09:38,875

Everything is linked.

86

00:09:39,041 --> 00:09:41,375

Nothing is self-sufficient.

87

00:09:41,541 --> 00:09:43,666

Water and air are inseparable,

88

00:09:43,833 --> 00:09:48,166

united in life

and for our life on Earth.

89

00:09:48,333 --> 00:09:50,833

Sharing is everything.

90

00:10:09,416 --> 00:10:14,583

The green expanse through the clouds  
is the source of oxygen in the air.

91

00:10:14,916 --> 00:10:17,250

70% of this gas,

92

00:10:17,416 --> 00:10:20,333

without which our lungs  
cannot function,

93

00:10:20,500 --> 00:10:24,708

comes from the algae that tint  
the surface of the oceans.

94

00:10:26,291 --> 00:10:29,166

Our Earth relies on a balance,

95

00:10:29,333 --> 00:10:32,000

in which every being  
has a role to play

96

00:10:32,166 --> 00:10:35,916

and exists only through the existence  
of another being.

97

00:10:36,250 --> 00:10:41,250

A subtle, fragile harmony  
that is easily shattered.

98

00:10:48,791 --> 00:10:53,375

Thus, corals are born  
from the marriage of algae and shells.

99

00:10:53,916 --> 00:10:57,666

Coral reefs cover  
less than 1% of the ocean floor,

100

00:10:57,833 --> 00:11:03,500

but they provide a habitat for thousands  
of species of fish, mollusks and algae.

101

00:11:03,666 --> 00:11:07,416

The equilibrium of every ocean



depends on them.

102

00:11:18,291 --> 00:11:22,625

The Earth counts time  
in billions of years.

103

00:11:22,791 --> 00:11:27,458

It took more than 4 billion years  
for it to make trees.

104

00:11:30,416 --> 00:11:34,375

In the chain of species,  
trees are a pinnacle,

105

00:11:34,541 --> 00:11:37,625

a perfect, living sculpture.

106

00:11:37,791 --> 00:11:39,666

Trees defy gravity.

107

00:11:39,833 --> 00:11:45,666

They are the only natural element  
in perpetual movement toward the sky.

108

00:11:45,833 --> 00:11:50,916

They grow unhurriedly toward the sun  
that nourishes their foliage.

109

00:12:03,458 --> 00:12:07,291

They have inherited  
from these miniscule cyanobacteria

110

00:12:07,458 --> 00:12:10,583

the power to capture light's energy.

111

00:12:10,958 --> 00:12:13,916

They store it and feed off it,

112

00:12:14,083 --> 00:12:16,708

turning it into wood and leaves,

113

00:12:16,875 --> 00:12:20,916

which then decompose  
into a mixture of water, mineral,

114

00:12:21,083 --> 00:12:24,000

vegetable and living matter.

115

00:12:27,666 --> 00:12:29,000

And so,

116

00:12:29,166 --> 00:12:30,625

gradually,

117

00:12:30,791 --> 00:12:32,791

soils are formed.

118

00:12:44,541 --> 00:12:48,666

Soils teem with the incessant activity  
of micro-organisms,

119

00:12:48,833 --> 00:12:53,166

feeding, digging,  
aerating and transforming.

120

00:12:53,333 --> 00:12:58,625

They make the humus, the fertile layer  
to which all life on land is linked.

121

00:13:20,041 --> 00:13:22,875

What do we know about life on Earth?

122

00:13:23,041 --> 00:13:27,000

How many species are we aware of?

A tenth of them?

123

00:13:27,166 --> 00:13:28,791

A hundredth perhaps?

124

00:13:28,958 --> 00:13:32,000

What do we know

about the bonds that link them?

125

00:13:43,166 --> 00:13:45,583

The Earth is a miracle.

126

00:13:45,750 --> 00:13:48,125

Life remains a mystery.

127

00:14:04,125 --> 00:14:08,250

Families of animals form,  
united by customs and rituals

128

00:14:08,416 --> 00:14:11,583

that are handed down  
through the generations.

129

00:14:30,791 --> 00:14:33,875

Some adapt  
to the nature of their pasture

130

00:14:34,041 --> 00:14:36,666

and their pasture adapts to them.

131

00:14:36,833 --> 00:14:38,833

And both gain.

132

00:14:39,000 --> 00:14:43,375

The animal sates its hunger  
and the tree can blossom again.

133

00:15:24,458 --> 00:15:27,125

In the great adventure

of life on Earth,

134

00:15:27,291 --> 00:15:29,375

every species has a role to play,

135

00:15:29,541 --> 00:15:32,541

every species has its place.

136

00:15:33,333 --> 00:15:36,250

None is futile or harmful.

137

00:15:36,416 --> 00:15:39,458

They all balance out.

138

00:15:52,375 --> 00:15:54,291

And that's where you,

139

00:15:54,458 --> 00:15:57,208

homo sapiens, wise human,

140

00:15:57,375 --> 00:15:59,291

enter the story.

141

00:16:01,375 --> 00:16:05,208

You benefit from a fabulous  
4-billion-year-old legacy

142

00:16:05,375 --> 00:16:07,250

bequeathed by the Earth.

143

00:16:14,708 --> 00:16:17,416

You are only 200,000 years old,

144

00:16:17,583 --> 00:16:20,666

but you have changed  
the face of the world.

145

00:16:22,458 --> 00:16:27,208

Despite your vulnerability, you have  
taken possession of every habitat

146

00:16:27,375 --> 00:16:32,125

and conquered swathes of territory,  
like no other species before you.

147

00:16:38,041 --> 00:16:41,166

After 180,000 nomadic years,

148

00:16:41,333 --> 00:16:43,750

and thanks to a more clement climate,

149

00:16:43,916 --> 00:16:45,958

humans settled down.

150

00:16:46,125 --> 00:16:49,333

They no longer depended  
on hunting for survival.

151

00:16:49,500 --> 00:16:53,166

They chose to live in wet environments  
that abounded in fish,

152

00:16:53,333 --> 00:16:55,291

game and wild plants.

153

00:16:55,458 --> 00:16:59,416

There where land,



water and life combine.

154

00:17:29,166 --> 00:17:30,375

Even today,

155

00:17:30,541 --> 00:17:34,625

the majority of humankind

lives on the continents' coastlines

156

00:17:34,791 --> 00:17:37,541

or the banks of rivers and lakes.

157

00:17:53,416 --> 00:17:56,583

Across the planet,

one person in four

158

00:17:56,750 --> 00:18:00,750

lives as humankind did

6,000 years ago,

159

00:18:00,916 --> 00:18:06,000

their only energy that which nature

provides season after season.

160

00:18:06,375 --> 00:18:10,250

It's the way of life  
of 1.5 billion people,

161

00:18:10,416 --> 00:18:15,208

more than the combined population  
of all the wealthy nations.

162

00:19:16,875 --> 00:19:21,791

But life expectancy is short  
and hard labor takes its toll.

163

00:19:21,958 --> 00:19:25,625

The uncertainties of nature  
weigh on daily life.

164

00:19:26,375 --> 00:19:29,666

Education is a rare privilege.

165

00:19:30,125 --> 00:19:33,375

Children are a family's only asset

166

00:19:33,541 --> 00:19:35,875

as long as every extra pair of hands

167

00:19:36,041 --> 00:19:39,333

is a necessary contribution

to its subsistence.

168

00:19:50,083 --> 00:19:51,791

Humanity's genius

169

00:19:51,958 --> 00:19:54,958

is to have always had a sense

of its weakness.

170

00:19:56,583 --> 00:20:01,541

The physical strength, with which  
nature insufficiently endowed humans,

171

00:20:01,708 --> 00:20:06,291

is found in animals that help them  
to discover new territories.

172

00:20:33,125 --> 00:20:36,875

But how can you conquer the world  
on an empty stomach?

173

00:20:41,041 --> 00:20:45,916

The invention of agriculture  
turned our history on end.

174

00:20:46,083 --> 00:20:49,250

It was less than 10,000 years ago.

175

00:20:49,416 --> 00:20:53,375

Agriculture  
was our first great revolution.

176

00:21:01,083 --> 00:21:03,458

It resulted in the first surpluses

177

00:21:03,625 --> 00:21:06,583

and gave birth to cities  
and civilizations.

178

00:21:13,708 --> 00:21:18,000

The memory of thousands of years  
scrabbling for food faded.

179

00:21:18,166 --> 00:21:22,833

Having made grain the yeast of life,  
we multiplied the number of varieties

180

00:21:23,000 --> 00:21:26,000

and learned to adapt them  
to our soils and climates.

181

00:21:44,041 --> 00:21:46,333

We are like every species on Earth.

182

00:21:46,500 --> 00:21:50,958

Our principal daily concern  
is to feed ourselves.

183

00:21:52,416 --> 00:21:54,625

When the soil is less than generous

184

00:21:54,791 --> 00:21:56,916

and water becomes scarce,

185

00:21:57,083 --> 00:22:00,083

we are able

to deploy prodigious efforts to extract

186

00:22:00,250 --> 00:22:03,333

from the land

enough to live on.

187

00:22:25,125 --> 00:22:29,541

Humans shaped the land with the patience

and devotion the Earth demands

188

00:22:29,708 --> 00:22:33,958

in an almost sacrificial ritual

performed over and over.

189

00:22:34,958 --> 00:22:40,416

Agriculture is still

the world's most widespread occupation.

190

00:22:40,791 --> 00:22:43,750

Half of humankind tills the soil,

191

00:22:44,791 --> 00:22:48,041

over three-quarters of them by hand.

192

00:22:54,958 --> 00:22:59,875

Agriculture is like a tradition handed  
down from generation to generation

193

00:23:00,041 --> 00:23:03,000

in sweat, graft and toil,

194

00:23:03,166 --> 00:23:07,958

because for humanity  
it is a prerequisite of survival.

195

00:23:16,333 --> 00:23:20,875

But after relying on muscle-power  
for so long, humankind found a way

196

00:23:21,041 --> 00:23:24,625

to tap into the energy  
buried deep in the Earth.

197

00:23:34,458 --> 00:23:38,541

These flames are also from plants.

A pocket of sunlight.

198

00:23:38,708 --> 00:23:41,541

Pure energy.

The energy of the sun,

199

00:23:41,708 --> 00:23:45,000

captured over millions of years

by millions of plants

200

00:23:45,166 --> 00:23:47,625

more than 100 million years ago.

201

00:23:47,791 --> 00:23:50,333

It's coal. It's gas.

202

00:23:50,500 --> 00:23:53,166

And, above all, it's oil.

203

00:24:07,833 --> 00:24:12,666



And this pocket of sunlight freed  
humans from their toil on the land.

204

00:24:14,166 --> 00:24:16,458

With oil began the era of humans

205

00:24:16,625 --> 00:24:19,791

who break free  
of the shackles of time.

206

00:24:20,166 --> 00:24:24,375

With oil, some of us  
acquired unprecedented comforts.

207

00:24:24,541 --> 00:24:27,083

And in 50 years, in a single lifetime,

208

00:24:28,041 --> 00:24:30,458

the Earth has been  
more radically changed

209

00:24:30,625 --> 00:24:34,416

than by all previous generations

of humanity.

210

00:24:40,250 --> 00:24:42,791

Faster and faster.

In the last 60 years,

211

00:24:42,958 --> 00:24:46,291

the Earth's population

has almost tripled.

212

00:24:46,458 --> 00:24:50,166

And over 2 billion people

have moved to the cities.

213

00:24:50,750 --> 00:24:52,625

Faster and faster.

214

00:24:52,791 --> 00:24:54,250

Shenzhen, in China,

215

00:24:54,416 --> 00:24:57,750

with hundreds of skyscrapers

and millions of inhabitants,

216

00:24:57,916 --> 00:25:02,458

was just a small fishing village  
barely 40 years ago.

217

00:25:02,833 --> 00:25:04,416

Faster and faster.

218

00:25:04,583 --> 00:25:07,583

In Shanghai,  
3,000 towers and skyscrapers

219

00:25:07,750 --> 00:25:12,041

have been built in 20 years.  
Hundreds more are under construction.

220

00:25:14,041 --> 00:25:18,041

Today, over half of the world's  
7 billion inhabitants

221

00:25:18,208 --> 00:25:19,833

live in cities.

222

00:25:33,083 --> 00:25:34,500

New York.

223

00:25:34,666 --> 00:25:36,666

The world's first megalopolis

224

00:25:36,833 --> 00:25:40,833

is the symbol of the exploitation  
of the energy the Earth supplies

225

00:25:41,000 --> 00:25:45,500

to human genius.

The manpower of millions of immigrants,

226

00:25:45,666 --> 00:25:50,000

the energy of coal,

the unbridled power of oil.

227

00:26:02,041 --> 00:26:05,583

America was the first

to harness the phenomenal,

228

00:26:05,750 --> 00:26:09,333

revolutionary power of "black gold".

229

00:26:11,458 --> 00:26:15,875

In the fields,

machines replaced men.

230

00:26:16,041 --> 00:26:18,875

A liter of oil

generates as much energy

231

00:26:19,041 --> 00:26:22,750

as 100 pairs of hands in 24 hours.

232

00:26:25,583 --> 00:26:29,125

In the United States,

only 3 million farmers are left.

233

00:26:29,666 --> 00:26:33,375

They produce enough grain

to feed 2 billion people.

234

00:26:35,250 --> 00:26:38,208

But most of that grain  
is not used to feed people.

235

00:26:38,375 --> 00:26:41,500

Here, and in all other  
industrialized nations,

236

00:26:41,666 --> 00:26:45,875

it is transformed into livestock feed  
or biofuels.

237

00:26:51,208 --> 00:26:55,333

The pocket of sunshine's energy  
chased away the specter of drought

238

00:26:55,500 --> 00:26:57,041

that stalked farmland.

239

00:26:57,750 --> 00:27:00,875

No spring escapes  
the demands of agriculture,

240

00:27:01,041 --> 00:27:05,750

which accounts for 70%  
of humanity's water consumption.

241

00:27:07,833 --> 00:27:10,833

In nature, everything is linked.

242

00:27:11,000 --> 00:27:14,500

The expansion of cultivated land  
and single-crop farming

243

00:27:14,666 --> 00:27:17,875

encouraged  
the development of parasites.

244

00:27:18,500 --> 00:27:21,916

Pesticides, another gift  
of the petrochemical revolution,

245

00:27:22,083 --> 00:27:23,416

exterminated them.

246

00:27:23,583 --> 00:27:27,375

Bad harvests and famine

became a distant memory.

247

00:27:27,541 --> 00:27:28,791

The biggest headache now

248

00:27:28,958 --> 00:27:33,416

was what to do with the surpluses  
engendered by modern agriculture.

249

00:27:34,458 --> 00:27:37,625

But toxic pesticides  
seeped into the air,

250

00:27:37,791 --> 00:27:40,875

soil, plants,  
animals, rivers and oceans.

251

00:27:41,458 --> 00:27:43,500

They penetrated the heart of cells

252

00:27:43,666 --> 00:27:48,083

similar to the mother cell  
shared by all forms of life.



253

00:27:48,541 --> 00:27:52,166

Are they harmful to the humans  
they released from hunger?

254

00:27:52,333 --> 00:27:56,000

These farmers  
in their yellow protective suits

255

00:27:56,166 --> 00:27:58,583

probably have a good idea.

256

00:28:07,083 --> 00:28:11,708

Then came fertilizers,  
another petrochemical discovery.

257

00:28:13,000 --> 00:28:18,291

They produced unprecedented results  
on plots of land thus far ignored.

258

00:28:24,416 --> 00:28:27,041

Crops adapted to soils and climates

259

00:28:27,208 --> 00:28:31,708

gave way to the most productive  
varieties and easiest to transport.

260

00:28:32,333 --> 00:28:34,000

And so, in the last century,

261

00:28:34,166 --> 00:28:37,625

three-quarters of the varieties  
developed by farmers

262

00:28:37,791 --> 00:28:41,333

over thousands of years  
have been wiped out.

263

00:28:45,708 --> 00:28:50,375

As far as the eye can see,  
fertilizer below, plastic on top.

264

00:28:50,541 --> 00:28:55,041

The greenhouses of Almeria, Spain,  
are Europe's vegetable garden.

265

00:28:55,208 --> 00:28:59,375

A city of uniformly sized vegetables  
waits every day

266

00:28:59,541 --> 00:29:03,333

for hundreds of trucks to take them  
to the continent's supermarkets.

267

00:29:06,791 --> 00:29:11,500

The more a country develops,  
the more meat its inhabitants consume.

268

00:29:12,041 --> 00:29:16,250

How can growing worldwide demand  
be satisfied without recourse

269

00:29:16,416 --> 00:29:19,416

to concentration camp-style  
cattle farms?

270

00:29:19,583 --> 00:29:21,250

Faster and faster.

271

00:29:21,416 --> 00:29:24,333

Like the life cycle of livestock,  
which may never see a meadow.

272

00:29:25,083 --> 00:29:30,250

Manufacturing meat faster than  
the animal has become a daily routine.

273

00:29:30,416 --> 00:29:34,250

In these vast foodlots,  
trampled by millions of cattle,

274

00:29:34,416 --> 00:29:37,041

not a blade of grass grows.

275

00:29:37,916 --> 00:29:42,291

A fleet of trucks from every corner  
of the country brings tons of grain,

276

00:29:42,458 --> 00:29:45,125

soy meal and protein-rich granules

277

00:29:45,291 --> 00:29:47,625

that will become tons of meat.

278

00:29:57,750 --> 00:30:00,708

The result is that

it takes 100 liters of water

279

00:30:00,875 --> 00:30:03,916

to produce 1 kilogram of potatoes,

280

00:30:04,083 --> 00:30:07,916

4,000 liters for 1 kilo of rice

281

00:30:08,083 --> 00:30:12,916

and 13,000 liters for 1 kilo of beef.

282

00:30:13,083 --> 00:30:17,833

Not to mention the oil guzzled

in the production process and transport.

283

00:30:23,041 --> 00:30:26,375

Our agriculture

has become oil-powered.

284

00:30:26,541 --> 00:30:29,000

It feeds

twice as many humans on Earth,

285

00:30:29,166 --> 00:30:32,833

but has replaced diversity

with standardization.

286

00:30:33,416 --> 00:30:36,875

It gives many of us comforts

we could only dream of,

287

00:30:37,041 --> 00:30:41,750

but it makes our way of life

totally dependent on oil.

288

00:30:42,166 --> 00:30:44,791

This is the new measure of time.

289

00:30:45,541 --> 00:30:50,541

Our world's clock now beats

to the rhythm of indefatigable machines

290

00:30:50,708 --> 00:30:53,541

tapping into the pocket of sunlight.

291

00:30:53,875 --> 00:30:57,416

The whole planet is attentive  
to these metronomes

292

00:30:57,583 --> 00:31:00,291

of our hopes and illusions.

293

00:31:00,916 --> 00:31:04,750

The same hopes and illusions  
that proliferate along with our needs,

294

00:31:04,916 --> 00:31:08,833

increasingly insatiable desires  
and profligacy.

295

00:31:09,000 --> 00:31:11,875

We know that the end of cheap oil  
is imminent,

296

00:31:12,041 --> 00:31:14,375

but we refuse to believe it.

297

00:31:16,083 --> 00:31:17,250

For many of us,

298

00:31:17,416 --> 00:31:21,541

the American dream is embodied  
by a legendary name.

299

00:31:21,708 --> 00:31:23,666

Los Angeles.

300

00:31:24,708 --> 00:31:27,833

In this city  
that stretches over 100 kilometers,

301

00:31:28,000 --> 00:31:32,416

the number of cars is almost equal  
to the number of inhabitants.

302

00:31:34,625 --> 00:31:39,208



Here, energy puts on a fantastic show  
every night.

303

00:31:51,083 --> 00:31:54,750

The days seem no more  
than a pale reflection of nights

304

00:31:54,916 --> 00:31:57,916

that turn the city into a starry sky.

305

00:32:07,083 --> 00:32:09,041

Faster and faster.

306

00:32:09,208 --> 00:32:13,291

Distances are no longer  
counted in miles, but in minutes.

307

00:32:13,458 --> 00:32:17,333

The automobile shapes new suburbs,  
where every home is a castle,

308

00:32:17,500 --> 00:32:20,791

a safe distance

from the asphyxiated city centers,

309

00:32:20,958 --> 00:32:25,375

and where neat rows of houses  
huddle around dead-end streets.

310

00:32:26,250 --> 00:32:28,458

The model of a lucky-few countries

311

00:32:28,625 --> 00:32:33,833

has become a universal dream  
preached by TVs all over the world.

312

00:32:34,000 --> 00:32:35,625

Even here in Beijing,

313

00:32:35,791 --> 00:32:40,041

it is cloned, copied and reproduced  
in these formatted houses

314

00:32:40,208 --> 00:32:43,583

that have wiped pagodas off the map.

315

00:32:50,333 --> 00:32:55,208

The automobile has become the symbol  
of comfort and progress.

316

00:32:56,416 --> 00:32:59,208

If this model were followed  
by every society,

317

00:32:59,375 --> 00:33:04,000

the planet wouldn't have 900 million  
vehicles, as it does today,

318

00:33:04,166 --> 00:33:06,958

but 5 billion.

319

00:33:07,416 --> 00:33:09,583

Faster and faster.

320

00:33:09,750 --> 00:33:13,291

The more the world develops,  
the greater its thirst for energy.

321

00:33:13,458 --> 00:33:17,791

Everywhere, machines dig, bore  
and rip from the Earth

322

00:33:17,958 --> 00:33:21,333

the pieces of stars buried  
in its depths since its creation...

323

00:33:22,333 --> 00:33:23,833

Minerals.

324

00:33:48,666 --> 00:33:52,750

As a privilege of power,  
80% of this mineral wealth

325

00:33:52,916 --> 00:33:57,125

is consumed  
by 20% of the world's population.

326

00:34:01,083 --> 00:34:03,083

Before the end of this century,

327

00:34:03,250 --> 00:34:08,250

excessive mining will have exhausted  
nearly all the planet's reserves.

328

00:34:26,250 --> 00:34:28,291

Faster and faster.

329

00:34:28,458 --> 00:34:33,125

Shipyards churn out oil tankers,  
container ships and gas tankers

330

00:34:33,291 --> 00:34:37,666

to cater for the demands  
of globalized industrial production.

331

00:34:37,833 --> 00:34:40,750

Most consumer goods travel  
thousands of kilometers

332

00:34:40,916 --> 00:34:45,000

from the country of production  
to the country of consumption.

333

00:34:46,166 --> 00:34:52,083

Since 1950, the volume of international trade has increased 20 times over.

334

00:35:00,125 --> 00:35:03,541

90% of trade goes by sea.

335

00:35:03,708 --> 00:35:09,000

500 million containers  
are transported every year.

336

00:35:09,166 --> 00:35:12,291

Headed for the world's major hubs  
of consumption,

337

00:35:12,458 --> 00:35:14,375

such as Dubai.

338

00:35:14,541 --> 00:35:18,000

Dubai is a sort of culmination  
of the Western model,

339

00:35:18,166 --> 00:35:21,166

a country where the impossible

becomes possible.

340

00:35:21,333 --> 00:35:24,000

Building artificial islands in the sea,  
for example.

341

00:35:27,625 --> 00:35:30,500

Dubai has few natural resources,

342

00:35:31,291 --> 00:35:35,458

but with oil money it can bring in  
millions of tons of material

343

00:35:36,041 --> 00:35:38,875

and workers from all over the planet.

344

00:35:39,791 --> 00:35:42,458

Dubai has no farmland,  
but it can import food.

345

00:35:45,250 --> 00:35:50,166

Dubai has no water, but it can afford  
to expend immense amounts of energy

346

00:35:50,333 --> 00:35:55,125

to desalinate seawater and build  
the world's highest skyscrapers.

347

00:35:55,291 --> 00:35:59,833

Dubai has endless sun,  
but no solar panels.

348

00:36:00,958 --> 00:36:06,291

It is the totem to total modernity  
that never fails to amaze the world.

349

00:36:17,541 --> 00:36:22,041

Dubai is like the new beacon  
for all the world's money.

350

00:36:30,083 --> 00:36:33,166

Nothing seems further removed  
from nature than Dubai,

351

00:36:33,333 --> 00:36:37,250

although nothing depends on nature



more than Dubai.

352

00:36:37,416 --> 00:36:41,416

Dubai is a sort of culmination  
of the Western model.

353

00:36:41,625 --> 00:36:46,666

We haven't understood that  
we're depleting what nature provides.

354

00:37:54,416 --> 00:37:59,250

Since 1950, fishing catches  
have increased fivefold

355

00:37:59,416 --> 00:38:03,791

from 18 to 100 million metric tons  
a year.

356

00:38:04,208 --> 00:38:08,833

Thousands of factory ships  
are emptying the oceans.

357

00:38:09,208 --> 00:38:12,041

Three-quarters of fishing grounds  
are exhausted,

358

00:38:12,208 --> 00:38:15,833

depleted or in danger of being so.

359

00:38:16,083 --> 00:38:19,041

Most large fish have been fished  
out of existence

360

00:38:19,208 --> 00:38:22,416

since they have no time to reproduce.

361

00:38:34,708 --> 00:38:39,458

We are destroying the cycle of a life  
that was given to us.

362

00:38:56,958 --> 00:39:02,166

At the current rate, all fish stocks  
are threatened with exhaustion.

363

00:39:03,375 --> 00:39:08,333

Fish is the staple diet

of one in five humans.

364

00:39:24,708 --> 00:39:28,458

We have forgotten

that resources are scarce.

365

00:39:29,291 --> 00:39:33,041

500 million humans

live in the world's desert lands,

366

00:39:33,208 --> 00:39:36,208

more than the combined population  
of Europe.

367

00:39:37,291 --> 00:39:38,958

They know the value of water.

368

00:39:39,125 --> 00:39:41,458

They know how to use it sparingly.

369

00:39:42,041 --> 00:39:46,000

Here, they depend on wells  
replenished by fossil water,

370

00:39:46,166 --> 00:39:50,666

which accumulated underground  
back when it rained on these deserts.

371

00:39:50,833 --> 00:39:53,458

25,000 years ago.

372

00:39:57,708 --> 00:40:01,250

Fossil water also enables crops  
to be grown in the desert

373

00:40:01,416 --> 00:40:04,333

to provide food for local populations.

374

00:40:04,500 --> 00:40:06,833

The fields' circular shape derives

375

00:40:07,000 --> 00:40:10,750

from the pipes that irrigate them  
around a central pivot.

376

00:40:12,125 --> 00:40:13,958

But there is a heavy price to pay.

377

00:40:14,791 --> 00:40:18,958

Fossil water

is a non-renewable resource.

378

00:40:30,250 --> 00:40:31,625

In Saudi Arabia,

379

00:40:31,791 --> 00:40:35,666

the dream of industrial farming  
in the desert has faded.

380

00:40:36,666 --> 00:40:38,625

As if on a parchment map,

381

00:40:38,791 --> 00:40:43,125

the light spots on this patchwork  
show abandoned plots.

382

00:40:43,291 --> 00:40:45,666

The irrigation equipment

is still there.

383

00:40:45,833 --> 00:40:48,541

The energy to pump water also.

384

00:40:48,708 --> 00:40:52,666

But the fossil water reserves  
are severely depleted.

385

00:41:00,333 --> 00:41:03,708

Israel turned the desert  
into arable land.

386

00:41:08,708 --> 00:41:12,500

Even though these hothouses  
are now irrigated drop by drop,

387

00:41:12,666 --> 00:41:17,000

water consumption continues  
to increase along with exports.

388

00:41:22,416 --> 00:41:26,041

The once mighty River Jordan

is now just a trickle.

389

00:41:26,208 --> 00:41:29,500

Its water has flown to supermarkets  
all over the world

390

00:41:29,666 --> 00:41:32,125

in crates of fruit and vegetables.

391

00:41:48,125 --> 00:41:50,666

The Jordan's fate is not unique.

392

00:41:50,833 --> 00:41:54,625

Across the planet,  
one major river in ten

393

00:41:54,791 --> 00:41:58,583

no longer flows into the sea  
for several months of the year.

394

00:42:08,708 --> 00:42:11,041

Deprived of the Jordan's water,

395

00:42:11,208 --> 00:42:16,041

the level of the Dead Sea goes down  
by over one meter per year.

396

00:42:39,583 --> 00:42:42,416

India risks being the country  
that suffers most

397

00:42:42,583 --> 00:42:45,333

from lack of water  
in the coming century.

398

00:42:45,500 --> 00:42:48,708

Massive irrigation  
has fed the growing population

399

00:42:49,458 --> 00:42:54,750

and in the last 50 years,  
21 million wells have been dug.

400

00:42:54,916 --> 00:42:56,916

In many parts of the country,



401

00:42:57,083 --> 00:43:00,708

the drill has to sink every deeper  
to hit water.

402

00:43:01,416 --> 00:43:06,458

In western India,  
30% of wells have been abandoned.

403

00:43:07,291 --> 00:43:10,916

The underground aquifers  
are drying out.

404

00:43:14,416 --> 00:43:19,333

Vast reservoirs will catch monsoon rains  
to replenish the aquifers.

405

00:43:22,125 --> 00:43:27,125

In the dry season, local village women  
dig them with their bare hands.

406

00:43:46,291 --> 00:43:48,250

Thousands of kilometers away,

407

00:43:48,416 --> 00:43:51,750

800 to 1,000 liters of water  
are consumed

408

00:43:51,916 --> 00:43:54,291

per person per day.

409

00:43:54,458 --> 00:43:57,666

Las Vegas was built out of the desert.

410

00:43:57,833 --> 00:44:00,125

Millions of people live there.

411

00:44:00,291 --> 00:44:02,833

Thousands more arrive every month.

412

00:44:03,000 --> 00:44:07,458

Its inhabitants are among the biggest  
water consumers in the world.

413

00:44:18,916 --> 00:44:22,583

Palm Springs is another desert city

with tropical vegetation

414

00:44:22,750 --> 00:44:24,708

and lush golf courses.

415

00:44:27,875 --> 00:44:31,625

How long can this mirage

continue to prosper?

416

00:44:35,583 --> 00:44:38,375

The Earth cannot keep up.

417

00:44:39,333 --> 00:44:42,541

The Colorado River,

which brings water to these cities,

418

00:44:42,708 --> 00:44:45,708

is one of those rivers

that no longer reaches the sea.

419

00:44:47,333 --> 00:44:52,416

Water levels in the catchment lakes

along its course are plummeting.

420

00:45:03,791 --> 00:45:09,666

Water shortages could affect nearly  
2 billion people before 2025.

421

00:45:36,333 --> 00:45:40,625

The wetlands represent  
6% of the surface of the planet.

422

00:45:41,083 --> 00:45:44,375

Under their calm waters  
lies a veritable factory,

423

00:45:44,541 --> 00:45:48,500

where plants and micro-organisms  
patiently filter the water

424

00:45:48,666 --> 00:45:51,125

and digest all the pollution.

425

00:45:52,083 --> 00:45:56,083

These marshes are indispensable  
environments for the regeneration

426

00:45:56,250 --> 00:45:58,833

and purification of water.

427

00:45:59,000 --> 00:46:02,916

They are sponges

that regulate the flow of water.

428

00:46:03,083 --> 00:46:04,958

They absorb it in the wet season

429

00:46:05,125 --> 00:46:07,625

and release it in the dry season.

430

00:46:37,416 --> 00:46:39,666

In our race to conquer more land,

431

00:46:39,833 --> 00:46:43,291

we have reclaimed them

as pasture for livestock,

432

00:46:43,583 --> 00:46:46,916

or as land for agriculture or building.

433

00:46:48,708 --> 00:46:53,208

In the last century,  
half the world's marshes were drained.

434

00:46:53,541 --> 00:46:57,666

We know neither their richness  
nor their role.

435

00:47:03,375 --> 00:47:05,916

All living matter is linked.

436

00:47:06,083 --> 00:47:10,875

Water, air, soil, trees.

437

00:47:11,041 --> 00:47:15,666

The world's magic  
is right in front of our eyes.

438

00:47:26,208 --> 00:47:30,666

Trees breathe groundwater  
into the atmosphere as light mist.

439

00:47:30,833 --> 00:47:35,125

They form a canopy that alleviates  
the impact of heavy rains.

440

00:47:35,291 --> 00:47:39,500

The forests provide the humidity  
that is necessary for life.

441

00:47:42,083 --> 00:47:43,583

They store carbon,

442

00:47:43,750 --> 00:47:47,208

containing more  
than all the Earth's atmosphere.

443

00:47:48,000 --> 00:47:53,791

They are the cornerstone of the climatic  
balance on which we all depend.

444

00:48:02,250 --> 00:48:05,083

The primary forests provide a habitat

445

00:48:05,250 --> 00:48:08,416

for three-quarters  
of the planet's biodiversity,

446

00:48:08,750 --> 00:48:12,375

that is to say,  
of all life on Earth.

447

00:48:21,416 --> 00:48:24,916

These forests provide the remedies  
that cure us.

448

00:48:25,083 --> 00:48:30,125

The substances secreted by these plants  
can be recognized by our bodies.

449

00:48:30,291 --> 00:48:33,041

Our cells talk the same language.

450

00:48:33,208 --> 00:48:36,125

We are of the same family.

451



00:48:54,583 --> 00:48:57,708

But in barely 40 years,  
the world's largest rainforest,

452

00:48:58,541 --> 00:49:02,833

the Amazon,  
has been reduced by 20%.

453

00:49:15,208 --> 00:49:19,083

The forest gives way to cattle ranches  
or soybean farms.

454

00:49:19,250 --> 00:49:23,666

95% of these soybeans are used  
to feed livestock and poultry

455

00:49:23,833 --> 00:49:25,583

in Europe and Asia.

456

00:49:25,750 --> 00:49:28,875

And so, a forest is turned into meat.

457

00:49:40,708 --> 00:49:44,208

Barely 20 years ago, Borneo,  
the 4th largest island

458

00:49:44,375 --> 00:49:48,375

in the world,  
was covered by a vast primary forest.

459

00:49:48,541 --> 00:49:50,875

At the current rate of deforestation,

460

00:49:51,041 --> 00:49:54,333

it will have disappeared  
within 10 years.

461

00:50:00,208 --> 00:50:04,041

Living matter  
bonds water, air, earth and the sun.

462

00:50:05,541 --> 00:50:08,708

In Borneo, this bond has been broken

463

00:50:08,875 --> 00:50:13,625

in what was one of the Earth's

greatest reservoirs of biodiversity.

464

00:50:23,291 --> 00:50:27,166

This catastrophe was provoked  
by the decision to produce palm oil,

465

00:50:27,333 --> 00:50:32,000

one of the most productive and consumed  
oils in the world, on Borneo.

466

00:50:32,625 --> 00:50:36,666

Palm oil not only caters  
to our growing demand for food,

467

00:50:36,833 --> 00:50:42,208

but also cosmetics, detergents  
and, increasingly, alternative fuels.

468

00:50:42,750 --> 00:50:47,791

The forest's diversity was replaced  
by a single species, the oil palm.

469

00:50:47,958 --> 00:50:50,583

For local people,  
it provides employment.

470

00:50:50,750 --> 00:50:53,666

It's an agricultural industry.

471

00:50:57,791 --> 00:51:01,750

Another example of massive deforestation  
is the eucalyptus.

472

00:51:01,916 --> 00:51:04,750

Eucalyptus is used to make paper pulp.

473

00:51:04,916 --> 00:51:08,208

Plantations are growing  
as demand for paper has increased

474

00:51:08,375 --> 00:51:10,750

fivefold in 50 years.

475

00:51:11,083 --> 00:51:14,625

One forest  
does not replace another forest.

476

00:51:15,041 --> 00:51:17,041

At the foot of these eucalyptus trees,

477

00:51:17,208 --> 00:51:22,416

nothing grows because their leaves form  
a toxic bed for most other plants.

478

00:51:25,500 --> 00:51:29,541

They grow quickly,  
but exhaust water reserves.

479

00:51:31,333 --> 00:51:33,583

Soybeans, palm oil,

480

00:51:33,750 --> 00:51:35,791

eucalyptus trees...

481

00:51:35,958 --> 00:51:41,041

Deforestation destroys the essential  
to produce the superfluous.

482

00:51:41,208 --> 00:51:42,458

But elsewhere,

483

00:51:42,833 --> 00:51:46,250

deforestation is a last resort  
to survive.

484

00:51:55,208 --> 00:51:57,041

Over 2 billion people,

485

00:51:57,875 --> 00:52:00,333

almost one third  
of the world's population,

486

00:52:00,500 --> 00:52:03,250

still depend on charcoal.

487

00:52:04,041 --> 00:52:05,208

In Haiti,

488

00:52:05,375 --> 00:52:07,666

one of the world's poorest countries,

489

00:52:07,833 --> 00:52:11,416

charcoal is one of the population's  
main consumables.

490

00:52:13,291 --> 00:52:15,750

Once the "pearl of the Caribbean",

491

00:52:15,916 --> 00:52:20,541

Haiti can no longer feed  
its population without foreign aid.

492

00:52:24,458 --> 00:52:29,000

On the hills of Haiti,  
only 2% of the forests are left.

493

00:52:30,541 --> 00:52:32,333

Stripped bare,

494

00:52:32,500 --> 00:52:34,333

nothing holds the soils back.

495

00:52:34,916 --> 00:52:38,875

The rainwater washes them  
down the hillsides as far as the sea.

496

00:52:39,041 --> 00:52:42,916

What's left is increasingly  
unsuitable for agriculture.

497

00:52:50,375 --> 00:52:54,833

In some parts of Madagascar,  
the erosion is spectacular.

498

00:52:55,208 --> 00:52:59,791

Whole hillsides bear deep gashes  
hundreds of meters wide.

499

00:53:00,375 --> 00:53:04,166

Thin and fragile,  
soil is made by living matter.

500

00:53:04,708 --> 00:53:07,500

With erosion,  
the fine layer of humus,

501



00:53:07,666 --> 00:53:11,583

which took thousands of years to form,  
disappears.

502

00:53:38,458 --> 00:53:41,625

Here's one theory of the story  
of the Rapanui,

503

00:53:41,791 --> 00:53:43,750

the inhabitants of Easter Island,

504

00:53:43,916 --> 00:53:46,625

that could perhaps  
give us pause for thought.

505

00:53:47,125 --> 00:53:50,166

Living on the most isolated island  
in the world,

506

00:53:50,333 --> 00:53:54,750

the Rapanui exploited their resources  
until there was nothing left.

507

00:53:55,250 --> 00:53:58,500

Their civilization did not survive.

508

00:53:58,958 --> 00:54:02,833

On these lands stood  
the highest palm trees in the world.

509

00:54:03,000 --> 00:54:04,750

They have disappeared.

510

00:54:04,916 --> 00:54:07,875

The Rapanui  
chopped them all down for lumber.

511

00:54:08,041 --> 00:54:10,333

They then faced  
widespread soil erosion.

512

00:54:13,375 --> 00:54:18,583

The Rapanui could no longer go fishing.  
There were no trees to build canoes.

513

00:54:24,166 --> 00:54:29,000

Yet the Rapanui formed one of the most  
brilliant civilizations in the Pacific.

514

00:54:29,166 --> 00:54:33,125

Innovative farmers, sculptors,  
exceptional navigators,

515

00:54:33,291 --> 00:54:37,833

they were caught in the vise of  
overpopulation and dwindling resources.

516

00:54:38,791 --> 00:54:43,250

They experienced social unrest,  
revolts and famine.

517

00:54:43,416 --> 00:54:46,791

Many did not survive the cataclysm.

518

00:55:06,000 --> 00:55:11,000

The real mystery of Easter Island is not  
how its strange statues got there,

519

00:55:11,166 --> 00:55:12,541

we know now.

520

00:55:12,708 --> 00:55:16,083

It is why the Rapanui  
didn't react in time.

521

00:55:23,791 --> 00:55:29,250

It's only one of a number of theories,  
but it has particular relevance today.

522

00:55:47,166 --> 00:55:52,166

Since 1950, the world's population  
has almost tripled.

523

00:55:52,333 --> 00:55:53,875

And since 1950,

524

00:55:54,041 --> 00:55:57,708

we have more fundamentally  
altered our island, the Earth,

525

00:55:57,875 --> 00:56:02,166

than in all

of our 200,000-year history.

526

00:56:02,333 --> 00:56:06,125

Nigeria is the biggest oil exporter  
in Africa,

527

00:56:06,291 --> 00:56:11,041

yet 70% of the population  
lives under the poverty line.

528

00:56:12,000 --> 00:56:17,125

The wealth is there, but the country's  
inhabitants don't have access to it.

529

00:56:17,291 --> 00:56:19,583

The same is true all over the globe.

530

00:56:19,750 --> 00:56:24,208

Half the world's poor  
live in resource-rich countries.

531

00:56:28,500 --> 00:56:32,666

Our mode of development

has not fulfilled its promises.

532

00:56:32,833 --> 00:56:38,708

In 50 years, the gap between rich  
and poor has grown wider than ever.

533

00:56:39,375 --> 00:56:40,375

Today,

534

00:56:40,541 --> 00:56:46,708

half the world's wealth is in the hands  
of the richest 2% of the population.

535

00:56:53,958 --> 00:56:56,458

Can such disparities be maintained?

536

00:56:57,208 --> 00:56:59,708

They are the cause  
of population movements

537

00:56:59,875 --> 00:57:02,833

whose scale we have yet  
to fully realize.

538

00:57:03,250 --> 00:57:08,416

The city of Lagos

had a population of 700,000 in 1960.

539

00:57:09,083 --> 00:57:13,666

That will rise to 16 million by 2025.

540

00:57:13,833 --> 00:57:18,000

Lagos is one of the fastest growing  
megapolises in the world.

541

00:57:18,166 --> 00:57:21,708

The new arrivals are mostly farmers  
forced off the land

542

00:57:21,875 --> 00:57:26,708

for economic or demographic reasons,  
or because of diminishing resources.

543

00:57:26,875 --> 00:57:29,833

This is a radically new type  
of urban growth,

544

00:57:30,000 --> 00:57:34,291

driven by the urge to survive  
rather than to prosper.

545

00:57:39,541 --> 00:57:45,333

Every week, over a million people swell  
the populations of the world's cities.

546

00:57:50,458 --> 00:57:56,291

1 human in 6 now lives in a precarious,  
unhealthy, overpopulated environment

547

00:57:56,458 --> 00:58:02,250

without access to daily necessities,  
such as water, sanitation, electricity.

548

00:58:37,083 --> 00:58:39,125

Hunger is spreading once more.

549

00:58:39,291 --> 00:58:42,500

It affects nearly 1 billion people.



550

00:59:14,416 --> 00:59:19,208

All over the planet, the poorest  
scrabble to survive, while we continue

551

00:59:19,375 --> 00:59:23,083

to dig for resources  
that we can no longer live without.

552

00:59:23,250 --> 00:59:25,458

We look farther and farther afield

553

00:59:25,625 --> 00:59:28,041

in previously unspoilt territory

554

00:59:28,208 --> 00:59:32,500

and in regions that are  
increasingly difficult to exploit.

555

00:59:39,125 --> 00:59:41,250

We're not changing our model.

556

00:59:42,250 --> 00:59:44,500

Oil might run out?

557

00:59:44,666 --> 00:59:48,500

We can still extract oil  
from the tar sands of Canada.

558

00:59:48,666 --> 00:59:53,500

The biggest trucks in the world  
move thousands of tons of sand.

559

00:59:53,833 --> 00:59:57,166

The process of heating  
and separating bitumen from the sand

560

00:59:57,333 --> 01:00:00,583

requires millions  
of cubic meters of water.

561

01:00:00,750 --> 01:00:03,166

Colossal amounts of energy are needed.

562

01:00:03,333 --> 01:00:06,041

The pollution is catastrophic.

563

01:00:06,208 --> 01:00:08,541

The most urgent priority, apparently,

564

01:00:08,708 --> 01:00:12,208

is to pick every pocket of sunlight.

565

01:00:45,291 --> 01:00:48,125

Our oil tankers

are getting bigger and bigger.

566

01:00:48,291 --> 01:00:51,541

Our energy requirements

are constantly increasing.

567

01:00:51,708 --> 01:00:54,791

We try to power growth

like a bottomless oven

568

01:00:54,958 --> 01:00:58,333

that demands more and more fuel.

569

01:01:14,166 --> 01:01:15,916

It's all about carbon.

570

01:01:16,083 --> 01:01:20,083

In a few decades, the carbon  
that made our atmosphere a furnace

571

01:01:20,250 --> 01:01:25,250

and that nature captured over millions  
of years, allowing life to develop,

572

01:01:25,416 --> 01:01:28,000

will have largely been pumped back out.

573

01:01:28,166 --> 01:01:31,500

The atmosphere is heating up.

574

01:01:31,666 --> 01:01:36,500

It would have been inconceivable for  
a boat to be here just a few years ago.

575

01:01:37,208 --> 01:01:41,583

Transport, industry,

deforestation, agriculture...

576

01:01:42,000 --> 01:01:46,125

Our activities release gigantic  
quantities of carbon dioxide.

577

01:01:46,791 --> 01:01:50,250

Without realizing it,  
molecule by molecule,

578

01:01:50,416 --> 01:01:53,916

we have upset  
the Earth's climatic balance.

579

01:01:57,416 --> 01:02:00,250

All eyes are on the poles,

580

01:02:01,166 --> 01:02:04,791

where the effects of global warming  
are most visible.

581

01:02:06,208 --> 01:02:08,541

It's happening fast, very fast.

582

01:02:09,125 --> 01:02:14,083

The north-west passage that connects  
America, Europe and Asia via the pole,

583

01:02:14,250 --> 01:02:15,666

is opening up.

584

01:02:16,333 --> 01:02:18,833

The arctic ice cap is melting.

585

01:02:21,583 --> 01:02:23,625

Under the effect of global warming,

586

01:02:23,791 --> 01:02:28,833

the ice cap has lost  
40% of its thickness in 40 years.

587

01:02:29,000 --> 01:02:33,708

Its surface area in the summer  
shrinks year by year.

588

01:02:34,583 --> 01:02:38,208

It could disappear  
in the summer months by 2030.

589

01:02:38,375 --> 01:02:41,000

Some say 2015.

590

01:02:52,791 --> 01:02:56,250

The sunbeams that the ice sheet  
previously reflected back

591

01:02:56,416 --> 01:02:59,916

now penetrate the dark water,  
heating it up.

592

01:03:00,083 --> 01:03:03,291

The warming process gathers pace.

593

01:03:11,916 --> 01:03:15,333

This ice contains the records  
of our planet.

594

01:03:15,500 --> 01:03:19,000

The concentration of carbon dioxide  
hasn't been so high

595

01:03:19,166 --> 01:03:21,875

for several hundred thousand years.

596

01:03:23,666 --> 01:03:27,958

Humanity has never lived  
in an atmosphere like this.

597

01:03:39,458 --> 01:03:44,791

Is excessive exploitation of resources  
threatening the lives of every species?

598

01:03:45,333 --> 01:03:46,750

Climate change

599

01:03:46,916 --> 01:03:48,500

accentuates the threat.

600

01:03:48,666 --> 01:03:52,375

By 2050,  
a quarter of the Earth's species



601

01:03:52,541 --> 01:03:55,125

could be threatened with extinction.

602

01:03:55,958 --> 01:03:57,500

In these polar regions,

603

01:03:57,666 --> 01:04:01,166

the balance of nature

has already been disrupted.

604

01:05:36,375 --> 01:05:37,708

Around the North Pole,

605

01:05:37,875 --> 01:05:43,250

the ice cap has lost 30%

of its surface area in 30 years.

606

01:05:44,666 --> 01:05:47,041

But as Greenland

rapidly becomes warmer,

607

01:05:47,208 --> 01:05:52,625

the freshwater of a whole continent  
flows into the salt water of the oceans.

608

01:06:04,291 --> 01:06:09,708

Greenland's ice contains 20%  
of the freshwater of the whole planet.

609

01:06:10,708 --> 01:06:15,583

If it melts,  
sea levels will rise by nearly 7 meters.

610

01:06:30,833 --> 01:06:33,666

But there is no industry here.

611

01:06:34,458 --> 01:06:37,375

Greenland's ice sheet suffers  
from greenhouse gases

612

01:06:37,541 --> 01:06:39,666

emitted elsewhere on Earth.

613

01:06:42,916 --> 01:06:46,166

Our ecosystem doesn't have borders.

614

01:06:46,333 --> 01:06:47,583

Wherever we are,

615

01:06:47,750 --> 01:06:51,166

our actions have repercussions  
on the whole Earth.

616

01:06:51,333 --> 01:06:55,375

Our planet's atmosphere  
is an indivisible whole.

617

01:06:55,541 --> 01:06:58,625

It is an asset we share.

618

01:07:01,416 --> 01:07:04,791

In Greenland,  
lakes are appearing on the landscape.

619

01:07:05,708 --> 01:07:10,166

The ice cap is melting at a speed  
even the most pessimistic scientists

620

01:07:10,333 --> 01:07:13,541

did not envision 10 years ago.

621

01:07:22,500 --> 01:07:26,166

More and more of these glacier-fed  
rivers are merging together

622

01:07:26,333 --> 01:07:28,666

and burrowing through the surface.

623

01:07:29,083 --> 01:07:33,041

It was thought the water would freeze  
in the depths of the ice.

624

01:07:33,208 --> 01:07:36,041

On the contrary,  
it flows under the ice,

625

01:07:36,208 --> 01:07:41,000

carrying the ice sheet into the sea,  
where it breaks into icebergs.

626

01:08:26,666 --> 01:08:29,250

As the freshwater  
of Greenland's ice sheet

627

01:08:29,416 --> 01:08:32,000

seeps into the salt water of the oceans,

628

01:08:32,166 --> 01:08:35,750

low-lying lands around the globe  
are threatened.

629

01:08:39,958 --> 01:08:41,958

Sea levels are rising.

630

01:08:42,125 --> 01:08:44,625

Water expanding as it gets warmer

631

01:08:44,791 --> 01:08:46,958

caused, in the 20th century alone,

632

01:08:47,125 --> 01:08:49,458

a rise of 20 centimeters.

633

01:08:50,041 --> 01:08:53,125

Everything becomes unstable.

634

01:08:53,291 --> 01:08:58,000

Coral reefs are extremely sensitive  
to the slightest change

635

01:08:58,166 --> 01:09:01,791

in water temperature.

30% have disappeared.

636

01:09:01,958 --> 01:09:06,208

They are an essential link  
in the chain of species.

637

01:09:10,291 --> 01:09:14,708

In the atmosphere, major wind streams  
are changing direction.

638

01:09:15,666 --> 01:09:18,541

Rain cycles are altered.

639

01:09:18,708 --> 01:09:21,958

The geography of climates is modified.

640

01:09:22,125 --> 01:09:24,666

The inhabitants of low-lying islands,

641

01:09:24,833 --> 01:09:28,125

here in the Maldives, for example,  
are on the front line.

642

01:09:28,750 --> 01:09:31,250

They are increasingly concerned.

643

01:09:31,416 --> 01:09:35,291

Some are already looking for new,  
more hospitable lands.

644

01:09:41,041 --> 01:09:44,666

If sea levels continue to rise  
faster and faster,

645

01:09:44,833 --> 01:09:49,541

what will major cities like Tokyo,  
the world's most populous city, do?

646

01:09:50,750 --> 01:09:55,125

Every year, scientists' predictions  
become more alarming.

647

01:09:57,083 --> 01:10:02,291

70% of the world's population  
lives on coastal plains.

648

01:10:02,458 --> 01:10:05,125

11 of the 15 biggest cities

649

01:10:05,291 --> 01:10:08,666

stand on a coastline or river estuary.

650

01:10:09,708 --> 01:10:13,083

As the seas rise,  
salt will invade the water table,

651

01:10:13,250 --> 01:10:16,500

depriving inhabitants



of drinking water.

652

01:10:16,958 --> 01:10:19,708

Migratory phenomena are inevitable.

653

01:10:19,875 --> 01:10:23,333

The only uncertainty  
concerns their scale.

654

01:10:54,375 --> 01:10:58,583

In Africa,  
Mount Kilimanjaro is unrecognizable.

655

01:10:59,208 --> 01:11:01,958

80% of its glaciers have disappeared.

656

01:11:02,125 --> 01:11:05,208

In summer,  
the rivers no longer flow.

657

01:11:05,375 --> 01:11:08,833

Local peoples are affected  
by the lack of water.

658

01:11:09,541 --> 01:11:14,083

Even on the world's highest peaks,  
in the heart of the Himalayas,

659

01:11:14,250 --> 01:11:17,625

eternal snows and glaciers  
are receding.

660

01:11:19,625 --> 01:11:23,458

Yet these glaciers play  
an essential role in the water cycle.

661

01:11:23,875 --> 01:11:26,916

They trap the water  
from the monsoons as ice

662

01:11:27,083 --> 01:11:30,208

and release it in the summer  
when the snows melt.

663

01:11:43,291 --> 01:11:47,541

The Himalayan glaciers are the source

of all the great Asian rivers,

664

01:11:47,708 --> 01:11:51,541

the Indus, Ganges,

Mekong, Yangtze Kiang...

665

01:11:52,458 --> 01:11:56,625

2 billion people depend on them

for drinking water

666

01:11:56,791 --> 01:12:00,625

and to irrigate their crops,

as in Bangladesh.

667

01:12:01,750 --> 01:12:04,833

On the delta

of the Ganges and Brahmaputra,

668

01:12:05,000 --> 01:12:09,500

Bangladesh is directly affected

by phenomena occurring in the Himalayas

669

01:12:09,666 --> 01:12:10,916

and at sea level.

670

01:12:11,375 --> 01:12:15,500

This is one of the most populous  
and poorest countries in the world.

671

01:12:15,666 --> 01:12:18,500

It is already hit by global warming.

672

01:12:18,833 --> 01:12:23,208

The combined impact of increasingly  
dramatic floods and hurricanes

673

01:12:23,375 --> 01:12:26,791

could make  
a third of its land mass disappear.

674

01:12:27,125 --> 01:12:30,958

When populations are subjected  
to these devastating phenomena,

675

01:12:31,125 --> 01:12:33,666

they eventually move away.

676

01:12:39,416 --> 01:12:41,916

Wealthy countries will not be spared.

677

01:12:42,083 --> 01:12:44,750

Droughts are occurring  
all over the planet.

678

01:12:44,916 --> 01:12:49,375

In Australia,  
half of farmland is already affected.

679

01:13:01,333 --> 01:13:05,500

We are in the process of compromising  
the climatic balance

680

01:13:05,666 --> 01:13:10,041

that has allowed us to develop  
over 12,000 years.

681

01:13:17,958 --> 01:13:22,166

More and more wildfires  
encroach on major cities.

682

01:13:23,291 --> 01:13:26,708

In turn,  
they exacerbate global warming.

683

01:13:26,875 --> 01:13:29,666

As the trees burn,  
they release carbon dioxide.

684

01:13:31,166 --> 01:13:36,000

The system that controls our climate  
has been severely disrupted.

685

01:13:36,166 --> 01:13:40,333

The elements on which it relies  
have been disrupted.

686

01:14:11,375 --> 01:14:16,291

The clock of climate change is ticking  
in these magnificent landscapes.

687

01:14:16,833 --> 01:14:20,250

Here in Siberia,

and elsewhere across the globe,

688

01:14:20,416 --> 01:14:24,041

it is so cold

that the ground is constantly frozen.

689

01:14:24,958 --> 01:14:27,875

It's known as permafrost.

690

01:14:28,041 --> 01:14:32,083

Under its surface

lies a climatic time-bomb.

691

01:14:32,250 --> 01:14:33,625

Methane,

692

01:14:33,791 --> 01:14:38,416

a greenhouse gas 20 times

more powerful than carbon dioxide.

693

01:14:53,416 --> 01:14:55,083

If the permafrost melts,

694

01:14:55,250 --> 01:14:58,583

the methane releases would cause  
the greenhouse effect

695

01:14:58,750 --> 01:15:03,000

to race out of control  
with consequences no one can predict.

696

01:15:08,416 --> 01:15:13,000

We would literally  
be in unknown territory.

697

01:15:20,625 --> 01:15:24,625

Humanity has no more than 10 years  
to reverse the trend

698

01:15:24,791 --> 01:15:27,750

and avoid  
crossing into this territory...

699

01:15:28,500 --> 01:15:31,500

Life on Earth  
as we have never known it.



700

01:15:59,958 --> 01:16:03,500

We have created phenomena  
we cannot control.

701

01:16:03,666 --> 01:16:05,000

Since our origins,

702

01:16:05,166 --> 01:16:09,750

water, air and forms of life  
are intimately linked.

703

01:16:11,875 --> 01:16:15,500

But recently  
we have broken those links.

704

01:16:17,000 --> 01:16:18,791

Let's face the facts.

705

01:16:18,958 --> 01:16:21,375

We must believe what we know.

706

01:16:24,333 --> 01:16:28,666

All we have just seen is a reflection  
of human behavior.

707

01:16:30,208 --> 01:16:33,458

We have shaped the Earth in our image.

708

01:16:34,041 --> 01:16:36,708

We have very little time to change.

709

01:16:37,291 --> 01:16:41,875

How can this century carry the burden  
of 9 billion human beings

710

01:16:42,041 --> 01:16:44,666

if we refuse to be called to account

711

01:16:44,833 --> 01:16:48,000

for everything we alone have done?

712

01:16:55,416 --> 01:17:01,000

20% of the world's population  
consumes 80% of its resources

713

01:17:20,708 --> 01:17:24,958

The world spends

12 times more on military expenditures

714

01:17:25,125 --> 01:17:27,791

than on aid to developing countries

715

01:17:40,708 --> 01:17:45,041

5,000 people a day die

because of dirty drinking water

716

01:17:45,208 --> 01:17:49,583

1 billion people

have no access to safe drinking water

717

01:18:00,458 --> 01:18:05,291

Nearly 1 billion people are going hungry

718

01:18:19,625 --> 01:18:24,625

Over 50% of grain

traded around the world

719

01:18:24,791 --> 01:18:29,083

is used for animal feed or biofuels

720

01:18:42,125 --> 01:18:47,083

40% of arable land

has suffered long-term damage

721

01:19:00,000 --> 01:19:04,500

Every year,

13 million hectares of forest disappear

722

01:19:16,416 --> 01:19:22,250

1 mammal in 4, 1 bird in 8, 1 amphibian

in 3 are threatened with extinction

723

01:19:22,416 --> 01:19:27,791

Species are dying out at a rhythm

1,000 times faster than the natural rate

724

01:19:37,583 --> 01:19:42,000

Three quarters of fishing grounds

are exhausted,

725

01:19:42,166 --> 01:19:44,916

depleted or in dangerous decline

726

01:19:54,750 --> 01:19:58,083

The average temperature

of the last 15 years

727

01:19:58,250 --> 01:20:00,666

has been the highest ever recorded

728

01:20:15,250 --> 01:20:20,708

The ice cap is 40% thinner

than 40 years ago

729

01:20:32,958 --> 01:20:41,000

There may be at least 200 million

climate refugees by 2050

730

01:20:48,708 --> 01:20:50,791

The cost of our actions is high.

731

01:20:50,958 --> 01:20:54,458

Others pay the price  
without having been actively involved.

732

01:20:54,916 --> 01:20:56,916

I have seen refugee camps

733

01:20:57,083 --> 01:21:00,041

as big as cities,  
sprawling in the desert.

734

01:21:00,625 --> 01:21:02,833

How many men,  
women and children

735

01:21:03,000 --> 01:21:05,083

will be left by the wayside tomorrow?

736

01:21:06,000 --> 01:21:09,875

Must we always build walls  
to break the chain of human solidarity,

737

01:21:10,041 --> 01:21:11,333

separate peoples

738

01:21:11,500 --> 01:21:15,166

and protect the happiness of some  
from others' misery?

739

01:21:15,333 --> 01:21:17,083

It's too late to be a pessimist.

740

01:21:17,833 --> 01:21:21,416

I know that a single human  
can knock down every wall.

741

01:21:21,583 --> 01:21:23,458

It's too late to be a pessimist.

742

01:21:23,625 --> 01:21:27,125

Worldwide,  
4 children out of 5 attend school.

743

01:21:27,291 --> 01:21:30,375

Never has learning been given  
to so many human beings.

744

01:21:30,541 --> 01:21:34,208

Everyone, from richest to poorest,  
can make a contribution.

745

01:21:34,375 --> 01:21:36,875

Lesotho,  
one of the world's poorest countries,

746

01:21:37,041 --> 01:21:41,541

is proportionally the one that invests  
most in its people's education.

747

01:21:41,708 --> 01:21:46,583

Qatar, one of the richest states,  
has opened up to the best universities.

748

01:21:46,750 --> 01:21:49,791

Culture, education,  
research and innovation

749

01:21:49,958 --> 01:21:52,166

are inexhaustible resources.



750

01:21:52,333 --> 01:21:54,083

In the face of misery and suffering,

751

01:21:54,250 --> 01:21:57,458

millions of NGOs prove that solidarity

752

01:21:57,625 --> 01:22:01,416

between peoples is stronger  
than the selfishness of nations.

753

01:22:01,958 --> 01:22:04,625

In Bangladesh,  
a man thought the unthinkable

754

01:22:04,791 --> 01:22:07,416

and founded a bank  
that lends only to the poor.

755

01:22:07,583 --> 01:22:12,666

In 30 years, it has changed  
the lives of 150 million people.

756

01:22:12,833 --> 01:22:16,583

Antarctica is a continent  
with immense natural resources

757

01:22:16,750 --> 01:22:19,125

that no country can claim for itself,

758

01:22:19,291 --> 01:22:22,916

a natural reserve  
devoted to peace and science.

759

01:22:23,083 --> 01:22:25,375

A treaty signed by 49 states

760

01:22:25,541 --> 01:22:28,458

has made it a treasure  
shared by all humanity.

761

01:22:28,625 --> 01:22:30,625

It's too late to be a pessimist.

762

01:22:30,791 --> 01:22:35,291

Governments have acted to protect

nearly 2% of territorial waters.

763

01:22:35,458 --> 01:22:39,083

It's not much but it's 2 times more  
than 10 years ago.

764

01:22:39,250 --> 01:22:42,875

The first natural parks were created  
just over a century ago.

765

01:22:43,041 --> 01:22:46,208

They cover over 13% of the continents.

766

01:22:46,375 --> 01:22:48,250

They create spaces  
where human activity

767

01:22:48,416 --> 01:22:53,041

is in step with the preservation  
of species, soils and landscapes.

768

01:22:53,208 --> 01:22:57,083

This harmony between humans and nature

can become the rule,

769

01:22:57,250 --> 01:22:59,041

no longer the exception.

770

01:22:59,208 --> 01:23:03,041

In the US, New York has realized  
what nature does for us.

771

01:23:03,208 --> 01:23:06,541

These forests and lakes  
supply all the city's drinking water.

772

01:23:07,666 --> 01:23:11,708

In South Korea,  
the forests had been devastated by war.

773

01:23:11,875 --> 01:23:14,541

Thanks to  
a national reforestation program,

774

01:23:14,708 --> 01:23:17,750

they once more cover

65% of the country.

775

01:23:17,916 --> 01:23:21,333

More than 75% of paper is recycled.

776

01:23:21,625 --> 01:23:26,791

Costa Rica has made a choice between  
military spending and land conservation.

777

01:23:26,958 --> 01:23:28,625

The country no longer has an army.

778

01:23:28,791 --> 01:23:32,416

It prefers to devote its resources  
to education, ecotourism

779

01:23:32,583 --> 01:23:35,083

and the protection  
of its primary forest.

780

01:23:35,250 --> 01:23:38,375

Gabon is one of the world's  
leading producers of wood.

781

01:23:38,541 --> 01:23:43,250

It enforces selective logging.

Not more than 1 tree every hectare.

782

01:23:43,416 --> 01:23:47,041

Its forests are one of the country's  
most important resources,

783

01:23:47,208 --> 01:23:49,583

but they have time to regenerate.

784

01:23:49,750 --> 01:23:53,541

Programs exist that guarantee  
sustainable forest management.

785

01:23:53,708 --> 01:23:55,666

They must become mandatory.

786

01:23:56,583 --> 01:24:00,625

For consumers and producers,  
justice is an opportunity to be seized.

787

01:24:01,458 --> 01:24:05,041

When trade is fair,  
when both buyer and seller benefit,

788

01:24:05,208 --> 01:24:08,791

everybody can prosper  
and earn a decent living.

789

01:24:09,625 --> 01:24:11,833

How can there be justice and equity

790

01:24:12,000 --> 01:24:15,125

between people  
whose only tools are their hands

791

01:24:15,291 --> 01:24:19,375

and those who harvest their crops  
with a machine and state subsidies?

792

01:24:22,625 --> 01:24:25,375

Let's be responsible consumers.

793

01:24:25,541 --> 01:24:27,791

Think about what we buy!

794

01:24:31,166 --> 01:24:32,958

It's too late to be a pessimist.

795

01:24:33,625 --> 01:24:36,375

I have seen agriculture

on a human scale.

796

01:24:36,541 --> 01:24:38,791

It can feed the whole planet

797

01:24:38,958 --> 01:24:42,875

if meat production doesn't take  
the food out of people's mouths.

798

01:24:43,500 --> 01:24:46,666

I have seen fishermen  
who take care what they catch

799

01:24:46,833 --> 01:24:49,375

and care for the riches of the ocean.



800

01:24:50,250 --> 01:24:52,958

I have seen houses  
producing their own energy.

801

01:24:53,125 --> 01:24:55,375

5,000 people live in the world's

802

01:24:55,541 --> 01:24:59,000

first ever eco-friendly district  
in Freiburg, Germany.

803

01:24:59,166 --> 01:25:01,291

Other cities partner the project.

804

01:25:01,458 --> 01:25:04,208

Mumbai is the thousandth to join them.

805

01:25:04,541 --> 01:25:09,333

The governments of New Zealand, Iceland,  
Austria, Sweden and other nations

806

01:25:09,500 --> 01:25:12,625

have made the development  
of renewable energy sources

807

01:25:12,791 --> 01:25:14,458

a top priority.

808

01:25:15,333 --> 01:25:20,541

80% of the energy we consume  
comes from fossil energy sources.

809

01:25:20,708 --> 01:25:21,541

Every week,

810

01:25:21,708 --> 01:25:25,958

two new coal-fired generating plants  
are built in China alone.

811

01:25:26,166 --> 01:25:30,375

But I have also seen, in Denmark,  
a prototype of a coal-fired plant

812

01:25:30,541 --> 01:25:34,250

that releases carbon into the soil  
rather than the air.

813

01:25:34,416 --> 01:25:37,416

A solution for the future?

Nobody knows yet.

814

01:25:37,583 --> 01:25:39,541

I have seen, in Iceland,

815

01:25:39,708 --> 01:25:42,333

an electricity plant

powered by the Earth's heat.

816

01:25:42,500 --> 01:25:44,000

Geothermal power.

817

01:25:44,208 --> 01:25:46,083

I have seen a sea snake

818

01:25:46,250 --> 01:25:49,375

lying on the swell

to absorb the energy of the waves

819

01:25:49,541 --> 01:25:51,458

and produce electricity.

820

01:25:52,333 --> 01:25:54,958

I have seen wind farms

off Denmark's coast

821

01:25:55,125 --> 01:25:58,375

that produce 20%

of the country's electricity.

822

01:25:58,541 --> 01:26:04,125

The USA, China, India, Germany

and Spain are the biggest investors

823

01:26:04,375 --> 01:26:05,958

in renewable energy.

824

01:26:06,125 --> 01:26:09,791

They have already created

over 2.5 million jobs.

825

01:26:10,291 --> 01:26:12,916

Where on earth

doesn't the wind blow?

826

01:26:14,041 --> 01:26:18,125

I have seen desert expanses

baking in the sun.

827

01:26:18,625 --> 01:26:21,000

Everything on Earth is linked,

828

01:26:21,166 --> 01:26:25,291

and the Earth is linked to the sun,

its original energy source.

829

01:26:25,458 --> 01:26:28,666

Can humans not imitate plants

and capture its energy?

830

01:26:29,416 --> 01:26:33,833

In one hour, the sun gives the Earth

the same amount of energy

831

01:26:34,000 --> 01:26:37,750

as that consumed  
by all humanity in one year.

832

01:26:37,916 --> 01:26:42,458

As long as the Earth exists,  
the sun's energy will be inexhaustible.

833

01:26:42,625 --> 01:26:43,625

All we have to do

834

01:26:43,791 --> 01:26:46,958

is stop drilling the Earth  
and start looking to the sky.

835

01:26:47,125 --> 01:26:50,208

All we have to do  
is learn to cultivate the sun.

836

01:26:50,375 --> 01:26:52,791

All these experiments  
are only examples,

837

01:26:52,958 --> 01:26:55,250

but they testify to a new awareness.

838

01:26:55,416 --> 01:26:58,583

They lay down markers  
for a new human adventure

839

01:26:58,750 --> 01:27:02,708

based on moderation,  
intelligence and sharing.

840

01:27:18,958 --> 01:27:21,416

It's time to come together.

841

01:27:22,333 --> 01:27:24,125

What's important

842

01:27:24,291 --> 01:27:26,125

is not what's gone,

843

01:27:26,291 --> 01:27:28,666

but what remains.

844

01:27:30,416 --> 01:27:33,375

We still have

half the world's forests,

845

01:27:33,541 --> 01:27:38,916

thousands of rivers, lakes and glaciers,

and thousands of thriving species.

846

01:27:42,208 --> 01:27:45,750

We know that the solutions

are there today.

847

01:27:46,833 --> 01:27:49,750

We all have the power to change.

848

01:27:51,458 --> 01:27:53,833

So what are we waiting for?

849

01:28:21,875 --> 01:28:26,041

It's up to us to write

what happens next



850

01:28:27,791 --> 01:28:31,250

Together

851

01:32:50,916 --> 01:32:55,458

get involved and join us on

[www.goodplanet.org](http://www.goodplanet.org)

852

01:32:56,666 --> 01:32:58,916

Special thanks to the 88.000 employees  
of the PPR Group

853

01:32:59,083 --> 01:33:00,750

for supporting the movie HOME